

AISHWARYAH RAVISANKAR(She/Her)

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PROFESSIONAL SUMMARY

Professional biostatistician with strong experience in clinical research, advanced statistical modeling, and regulatory-compliant trial analytics. Skilled in a range of methodologies for model validation, performance evaluation, parametric and non-parametric testing, frequentist and Bayesian approaches, and cross-platform biomarker comparisons. I have worked on close to 20 clinical trials including anesthetics, oncology, cardiovascular drugs, arthritis among other therapeutic areas.

EDUCATION

University of Illinois at Chicago (UIC)

May 2023

Master of Public Health (Biostatistics)

Saveetha Institute of Medical and Technical Sciences, Chennai, India

2019

Bachelor of Dental Surgery

SKILLS & ABILITIES

Languages: SAS | R | SPSS | SQL | Python

Tools: Tableau | PowerBI | REDCap | R studio | Dedoose

Statistical techniques: Generalized Linear Modeling | Mixed-Effects Models | Bayesian Modeling | Survival Analysis (Cox, Kaplan-Meier) | Principal Component Analysis (PCA) | Factor Analysis (FA) | Win-Ratio Analysis | ANCOVA | Parametric & Non-Parametric Testing | Effect Size Estimation (Cohen's d, Odds Ratios, Hazard Ratios) | Model Validation (K-Fold Cross-Validation, Bootstrapping, Monte Carlo) | Agreement Analyses (Bland-Altman, ICC, CCC)

Research: IRB Protocols | Manuscript Composition | Survey Creation | Clinical Trials | Research Design

Data Management: Database Design & Management | Missing Data Handling

Interpersonal Skills: Verbal & Written Communication | Problem Solving | Ability to Work Under Pressure | Organizational Leadership | Teamwork & Collaboration | Project Management | Analytical & Critical thinking | Mentoring & Oversight of Statistical Programming Teams

RELATED EXPERIENCE

Lead Biostatistician, Huntington Medical Research Institute.

November 2024 – Ongoing

Pasadena, California.

- Performed power analyses, sample size and effect size estimation using methods such as Cohen's d, odds ratios, hazard ratios, and partial eta-squared. Applied both parametric (ANCOVA, t-tests) and non-parametric tests (chi-square, Wilcoxon, Mann-Whitney U, Kruskal-Wallis). Led survival analysis (Cox, Kaplan-Meier) and platform comparison methods (Bland-Altman, ICC)
- Designed and executed comprehensive statistical analyses including model validation techniques (K-fold cross-validation, bootstrapping, Monte Carlo), frequentist and Bayesian models, supervised learning (regression, multi-level models, random forest), and unsupervised techniques (PCA, FA).
- Collaborated with investigators, research staff and core directors developed data management strategies using REDCap and R, ensured data quality, mentored students, authoring four manuscripts, and supported scientific reporting and presentations.

Lead Biostatistician, Pharmadata Associates.

July 2023 – November 2024

Irvine, California.

- Led clinical trial design and conducted sample size calculations and power analyses based on sponsor requests.
- Independently developed Statistical Analysis Plans (SAPs) and TLF shells across all three phases of clinical trials in adherence with CDISC standards.
- Programmed all Tables in the 14.2 series and created all associated figures for 20 clinical trials spanning therapeutic areas including anesthetics, oncology, cardiovascular drugs, arthritis, and bone-modifying agents using SAS and R programming language.
- Interpret results and address limitations in study design and/or statistical methods.
- Presented overviews of clinical trials to programming teams, tracked progress of dataset development, and ensured adherence to deadlines and contributed to development of SOPs and Working Instructions (WIs).

Biostatistics Intern, UIC, Chicago Cancer Centre.

June 2022 – May 2023

Chicago, Illinois.

- Built a REDCap database to manage student and mentor participation in the UI Cancer Center Summer Program, and conducted statistical analysis in R to evaluate publication outcomes.
- Integrated REDCap API with R to automate publication retrieval for program participants using PubMed queries, applied dimensionality reduction to identify predictors of publication success, and used Tableau to generate descriptive summaries by group and mentor involvement over multiple years.

Research Analyst, University of Chicago.

July 2022 – December 2022

Chicago, Illinois

- Performed data cleaning and analysis using SAS for participant-level research data related to genetic mutations and pediatric growth outcomes.
- Conducted linear regression analysis with assumption checks, adjusted for predictors identified from literature, and used a stepwise forward variable selection method. Created tables, charts, and visualizations in Tableau, and prepared statistical reports for manuscript development.